

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

This month, learn more about bending icicles in R&D, discover our planet's carbon budget in Earth, and we learn about the hunt for supersymmetry in Cutting Edge.



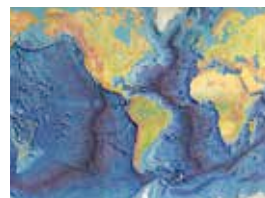
Black hole far side

Astronomers have observed the luminous x-ray echoes reflected from behind a black hole, by carefully looking for the differences with the luminous x-rays reaching us directly from the black hole. This is the first detection of light from behind a black hole. <https://dgit.in/blackhole>

WHAT'S NEW

Geologists measure the Earth's internal temperature

Surrounding the planet is a chain of volcanoes, oceanic trenches and subduction zones for continental plates, all caused by the process of plate tectonics. The section that spans the Pacific ocean is called the Ring of Fire, and contains within it volcanic belts, trench and ridges. Along the ridges, the temperatures are so hot that rock melts and rises upwards towards the surface through the crust, fracturing the ocean floor, and changing the surface of the planet. This erupted material forms what is known as "sea glass". Researchers used thousands of sea glass samples to analyse their chemical composi-



tion and figure out at what temperature the constituent elements melted, and the temperature at which sea glass was formed. The

readings provided an internal temperature check of the planet, used to make a global map of the ocean ridges. While the temperatures reached 1,350 degrees Celcius along most of the ridges, there were hotspots where the temperatures rose to 1,600 degrees Celsius. <https://dgit.in/seaglass>

Vegetation growth in the Northern Hemisphere stunted by emissions

Researchers have conducted a wide scale study of all vegetation growth in the Northern Hemisphere over 30 years and found that the growth has been stunted because of emissions. An increase in carbon dioxide in the atmosphere increased plant growth, but there is cause for concern as the increase in temperatures also increases the atmospheric



demand for water, increasing the number of extremely hot days and draughts, increasing the stress for the vegetation. The research also shows that increasing temperatures are going to inhibit vegetation growth even further, reducing

their ability of scrubbing out the carbon dioxide from the atmosphere. The water vegetation constraints discovered to be widespread across the Northern Hemisphere is a cause for concern, as these may shift the planet to a period of decreasing land carbon sink strength, from an increasing land carbon sink strength.

<https://dgit.in/treenogro>



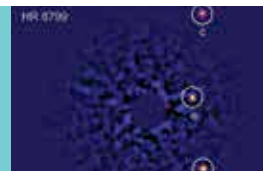
Reprogrammable satellite launched

The ESA has a flexible communications satellite with innovative software on board that can be used to completely repurpose the satellite after it has been deployed, meeting new emerging requirements as technology advances. After the mission, it will be parked in a safe orbit. <https://dgit.in/rsat>



Exoplanet moon formation

For the first time, astronomers have captured images of the circumplanetary satellite forming disc around a gas giant about 400 light years away. The observations allow researchers to see moon formation in action, as well as uncover the processes that lead to planet formation. <https://dgit.in/firstdisc>



Super Jupiter days measured

Astronomers have measured the rotational periods of four gas giants in HR 8799, which is famous for being the first extrasolar system to be directly imaged. Depending on the tilt of the planets, the days could be as short as 3 hours or as long as 24 hours. <https://dgit.in/HR8799>